

Literal V59 Source-Operator Two-Lane Block Attachment

Exact covariance and the external-copy norm obstruction

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August 25, 2026

Abstract

The V59 twin-prime Gate-B scalar was known as a signed four-packet remainder, while the recent covariance route required a literal two-lane coefficient attachment. We construct such an attachment directly on the finite physical source-index space. The complete prime weight, unit masks, deleted diagonal, kernel orientation, and real physical coefficients are packaged in one operator A_x , giving the exact identity $\mathfrak{C}_x = \langle w, A_x \beta \rangle$. Any declared disjoint source partition then gives an exactly-once ordered block decomposition. Tagged external copies turn the block sum into one Hilbert covariance. This positive result comes with an exact obstruction: for m input blocks the copied output lane has squared norm $m\|w\|^2$, and separating the input blocks can remove cancellation present in $A_x \beta$. Thus the physical-index attachment is exact but does not supply the norm-payable primitive-frequency attachment needed for the hard-window theorem. An exact-rational certificate independently replays every admissible triple and the two norm failures. No arithmetic L^2 estimate or twin-prime conclusion is claimed.

1 The source-attachment problem

The frozen V59 record reduces the open analytic gate to the scalar

$$\mathfrak{C}_x = \sum_{q \in \mathcal{Q}} q \sum_{\substack{t, u \in I_x \\ t \neq u, q \nmid tu}} \beta(t) w(u) K_H(u - t) u_1(u\bar{t}; q), \quad (1)$$

where $I_x = (x/2, x] \cap \mathbb{Z}$, q is prime, $u_1(a; q) = \mathbf{1}_{a \equiv 1 \pmod{q}} - 1/(q-1)$, and the physical β, w are real [1]. TPC-243 provides a strong bilinear transfer once two coefficient vectors have been identified in one separated-frequency synthesis space [2]. TPC-246 then gives sharp aggregate disk geometry, but explicitly assumes the relevant joint feasible family [3]. Neither theorem constructs the literal V59 coefficient lanes.

Our narrower question is algebraic: can (1) itself be written as a two-lane covariance without changing its physical coefficients? The answer is yes on the source-index space. The resulting representation also shows why existence is weaker than a usable hard-window attachment.

Conventions. All complex Hilbert inner products are conjugate-linear in the first slot. The physical sequence w is real. If it were replaced by a genuinely complex coefficient while the unconjugated factor $w(u)$ in (1) were retained, the first Hilbert lane would be $\bar{w}$.

2 Literal source operator

Let $\mathcal{H}_x = \mathbb{C}^{I_x}$. Define the finite matrix

$$A_x(u, t) = \mathbf{1}_{u \neq t} \sum_{q \in \mathcal{Q}} q \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} K_H(u - t) \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right). \quad (2)$$

The two unit masks make t^{-1} well defined modulo q , and $ut^{-1} \equiv 1$ exactly when $u \equiv t$. Consequently (2) is a repackaging of the full kernel in (1); it does not alter the outer q , diagonal deletion, or kernel orientation.

Proposition 1 (Literal two-lane source operator). *For the real physical V59 lanes,*

$$\mathfrak{C}_x = \langle w, A_x \beta \rangle_{\mathcal{H}_x}.$$

No Hermiticity or positivity of A_x is required.

Proof. Expand the right side. Conjugate-linearity in the first slot gives $\sum_u \overline{w(u)} \sum_t A_x(u, t) \beta(t)$. Since w is real, this is exactly (1) after the unit-residue equivalence above. $\square$

The proposition is source-native in a precise sense: both vector coordinates are the literal physical integers in I_x . It is not yet an identification with primitive rational frequencies, and A_x is not asserted to be a Gram matrix T^*T .

3 Exactly-once hard blocks

Fix a finite disjoint partition $I_x = \bigsqcup_{b \in \mathcal{B}} I_b$ and let P_b multiply by $\mathbf{1}_{I_b}$. Put

$$\beta_b = P_b \beta, \quad w_c = P_c w, \quad A_{cb} = P_c A_x P_b.$$

Theorem 1 (Exactly-once source blocks). *One has*

$$A_x = \sum_{b,c} A_{cb}, \quad \mathfrak{C}_x = \sum_{b,c} \langle w_c, A_{cb} \beta_b \rangle. \quad (3)$$

Every admissible triple (q, t, u) in (1) occurs in exactly one summand of (3).

Proof. The projections are self-adjoint, pairwise orthogonal, and sum to the identity. Hence

$$\sum_{b,c} A_{cb} = \left(\sum_c P_c \right) A_x \left(\sum_b P_b \right) = A_x.$$

Because $A_{cb} \beta_b \in P_c \mathcal{H}_x$, orthogonality of the output supports gives the scalar identity. Finally every t has one input label $b(t)$ and every u has one output label $c(u)$. The complete summand for an admissible (q, t, u) therefore lies in $(c(u), b(t))$ and nowhere else. $\square$

The disjoint hypothesis is essential for the literal “exactly once” wording. V59 also uses a smooth bounded-overlap partition of unity. That is an exact weighted pair partition, but one physical pair can occur in several weighted block pairs. We do not identify the two constructions.

4 Tagged covariance and its exact toll

For each ordered pair (b, c) introduce a tagged copy $\mathcal{H}_c^{(b,c)} \cong P_c \mathcal{H}_x$ and set

$$\mathcal{H}_{\text{ext}} = \bigoplus_{b,c} \mathcal{H}_c^{(b,c)}, \quad \mathbf{B} = \bigoplus_{b,c} A_{cb} \beta_b, \quad \mathbf{W} = \bigoplus_{b,c} w_c.$$

Theorem 2 (External-copy covariance and norm ledger). *If $m = |\mathcal{B}|$, then*

$$\langle \mathbf{W}, \mathbf{B} \rangle_{\mathcal{H}_{\text{ext}}} = \mathfrak{C}_x, \tag{4}$$

$$\|\mathbf{W}\|^2 = m \|w\|^2, \tag{5}$$

$$\|\mathbf{B}\|^2 = \sum_{b,c} \|A_{cb} \beta_b\|^2. \tag{6}$$

There is no general equality between (6) and $\|A_x \beta\|^2$.

Proof. Orthogonality of the tags turns each direct-sum inner product and norm into the sum of its coordinate values. Equation (4) is then Theorem 1. Every w_c appears once for each of the m input blocks, proving (5); (6) is immediate.

For the final assertion, use singleton blocks on three coordinates and

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \beta = (0, 1, 1)^T.$$

Then $A\beta = 0$, but the two nonzero tagged block outputs are $+1$ and -1 . Their squared direct-sum norm is 2. This is an exact counterexample. $\square$

The obstruction has two parts. The w lane pays a compulsory factor $\sqrt{m}$ if every ordered block pair is retained. The B lane moves cross-input-block cancellation out of the norm. Applying a near-isometry error to these lanes without paying both effects would therefore be invalid.

5 Certificate, route boundary, and conclusion

The certificate uses six physical indices, primes 5 and 7, three hard blocks, nonconstant real β, w , and the rational kernel $K(h) = 1/(1 + |h|)$. It reconstructs every matrix entry, compares the direct and block scalars, and counts each admissible (q, t, u) exactly once. The independent checker imports no producer code and rejects type, digest, coverage, norm, and arithmetic-promotion mutations. A separate stress test checks five partitions, the complex- w orientation, and the block-norm cancellation example.

The exact maximum claim is

`PROVED_STRUCTURAL_L1_LITERAL_V59_SOURCE_OPERATOR_ATTACHMENT_WITH_NORM_OBSTRUCTION.`

The primitive-frequency attachment, the TPC-243 synthesis map, a TPC-244 common multiplier, source product realizability, arithmetic L^2 , fixed-atom credit, strict 1/400, full Gate B, and the twin-prime conclusion remain open. The next minimal problem is no longer whether a physical two-lane pairing exists. It is to characterize the joint covariance family generated when the same output lane w_c is shared by all input-block probes.

References

- [1] Prime Dynamics Theory Project. V59 polarized local bdh scalar compiler, 2026. Committed repository research record.
- [2] Liang Wang. Hard-window near-isometry and signed bilinear transfer, 2026. TPC-243 committed repository paper.
- [3] Liang Wang. Weighted covariance-disk reassembly and hard-window margins, 2026. TPC-246 committed repository paper.